

An Interpretable Dual Graph Autoencoder Framework

for Autism Spectrum Disorder detection from resting-state fMRI

Georgia Kapetadimitri · Supervisor: Eftychios Protopapadakis

Department of Applied Informatics · University of Macedonia, Thessaloniki, Greece



THE IDEA IN ONE LINE

A classifier whose predictive features **are** the explanation — every decision traces back to specific anatomical brain connections.

01 The problem

Autism Spectrum Disorder diagnosis is still behaviour-based and time-intensive. Resting-state fMRI could complement it — regions as nodes, functional connections as edges — a natural fit for a graph. But an imaging tool only helps if its decisions are transparent enough for clinicians to trust.

Black-box

most fMRI classifiers give no clinical insight

Networks

ASD understood as a connectivity disorder

02 Goal & conclusion

THE GOAL

Build a classifier whose predictive features are themselves edge-level explanations — not bolted on after training.

THE CONCLUSION

A dual, class-specific graph autoencoder whose per-edge reconstruction-error features (ΔE) drive prediction **and** localise atypical connectivity — every decision traces to specific anatomical brain connections. The flagged regions align with established ASD neuroimaging.

03 Dataset — ABIDE

Autism Brain Imaging Data Exchange — a multi-site resting-state fMRI dataset.

862

subjects (after QC)

403

ASD

459

Control

111

ROIs (Harvard–Oxford)

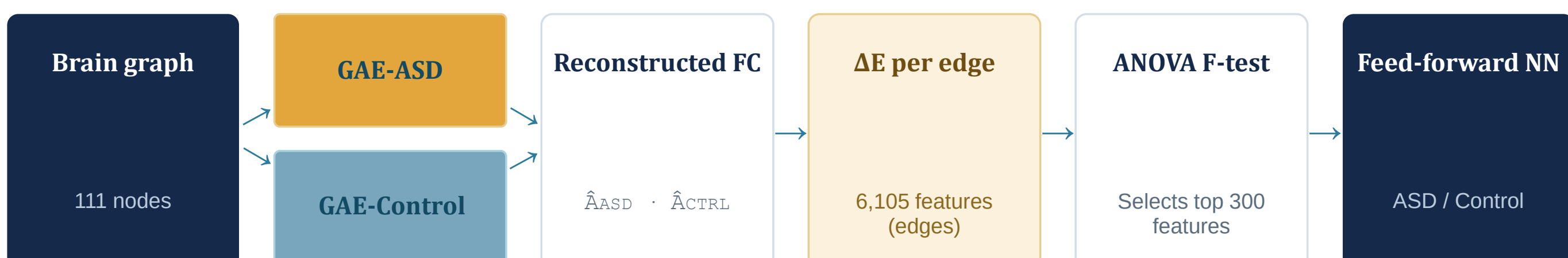
- Mean BOLD time-series per cortical + subcortical ROI
- 17 acquisition sites — multi-site heterogeneity is a known ABIDE difficulty driver

04 Building the brain graph

- Nodes (V):** 111 cortical + subcortical ROIs
- Node features (X):** each node's Pearson correlations with all 111 ROIs — a 111-dim vector
- Edges (E):** fully connected — every region pair
- Edge weights (w):** partial correlations of BOLD series — partial FC matrix $A \in \mathbb{R}^{111 \times 111}$
- 6,105** unique edges per subject (upper triangle of A)

05 Framework

Two class-specific autoencoders are trained separately — their reconstruction errors become the interpretable features.



06 Architecture

Input $G = (X, A)$, $X, A \in \mathbb{R}^{111 \times 111}$

Encoder — 3-layer GraphSAGE with residual connections:

WeightedSAGE · 64 · skip · LN · LeakyReLU

WeightedSAGE · 64 · skip · LN · LeakyReLU

WeightedSAGE · 64 · skip · LN · LeakyReLU

Embeddings $Z \in \mathbb{R}^{111 \times 64}$

Decoder — MLP on $[z_i || z_j] \rightarrow$ reconstructed $\hat{A}$

07 Margin-based specialization loss

Force each GAE to reconstruct only its own class — and to fail on the opposite class.

$$L_{\text{total}} = L_{\text{recon}}(\text{own}) + \alpha \cdot \max(0, m - L_{\text{recon}}(\text{opp}))$$

$m = 0.02$

margin

$\alpha = 1$

specialization weight

$\sigma = 0.02$

Gaussian edge augmentation

08 Interpretable feature: ΔE

Per edge, the difference between the two GAEs' reconstruction errors:

$$\Delta E = \text{Err}_{\text{ASD}} - \text{Err}_{\text{CTRL}}$$

$\Delta E > 0$

GAE-ASD struggles → ASD-atypical

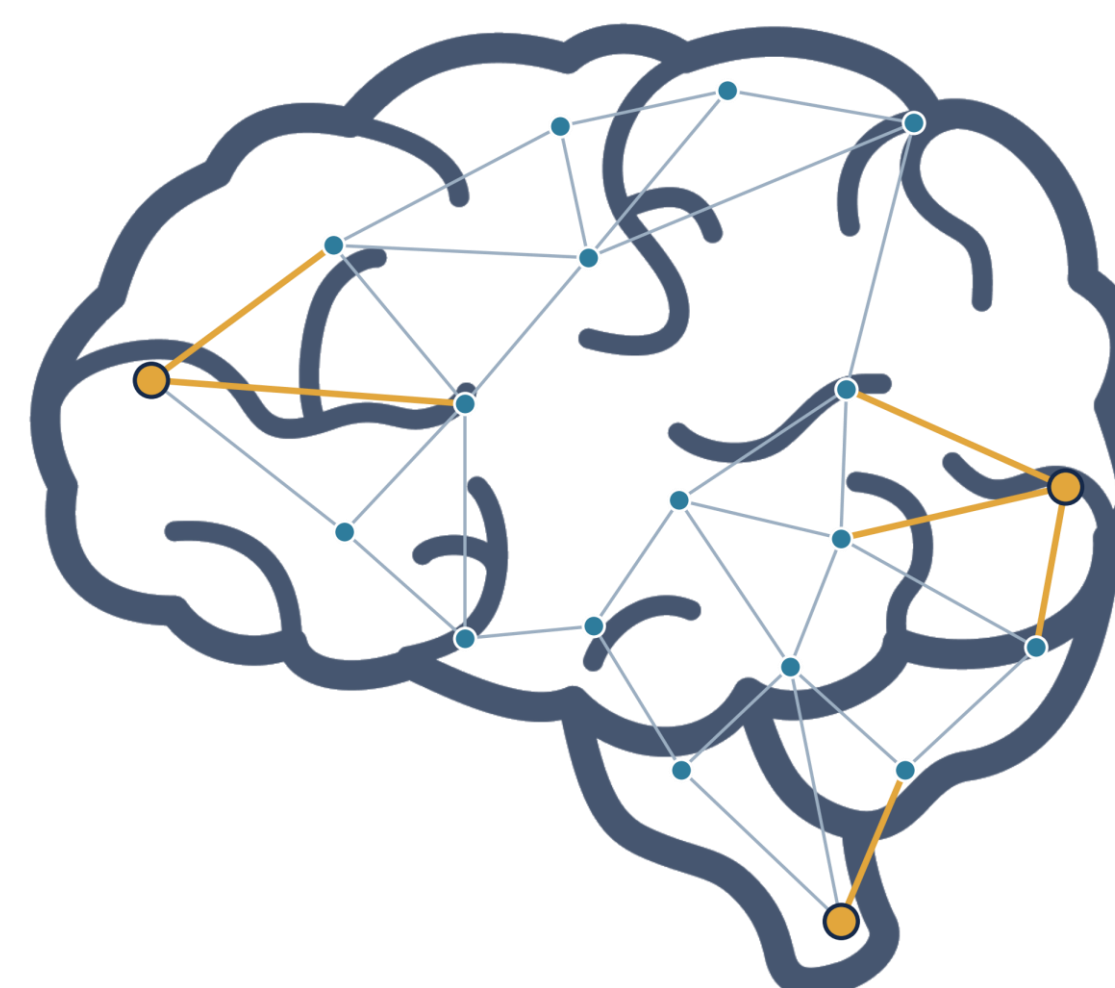
$\Delta E < 0$

GAE-CTRL struggles → control-atypical

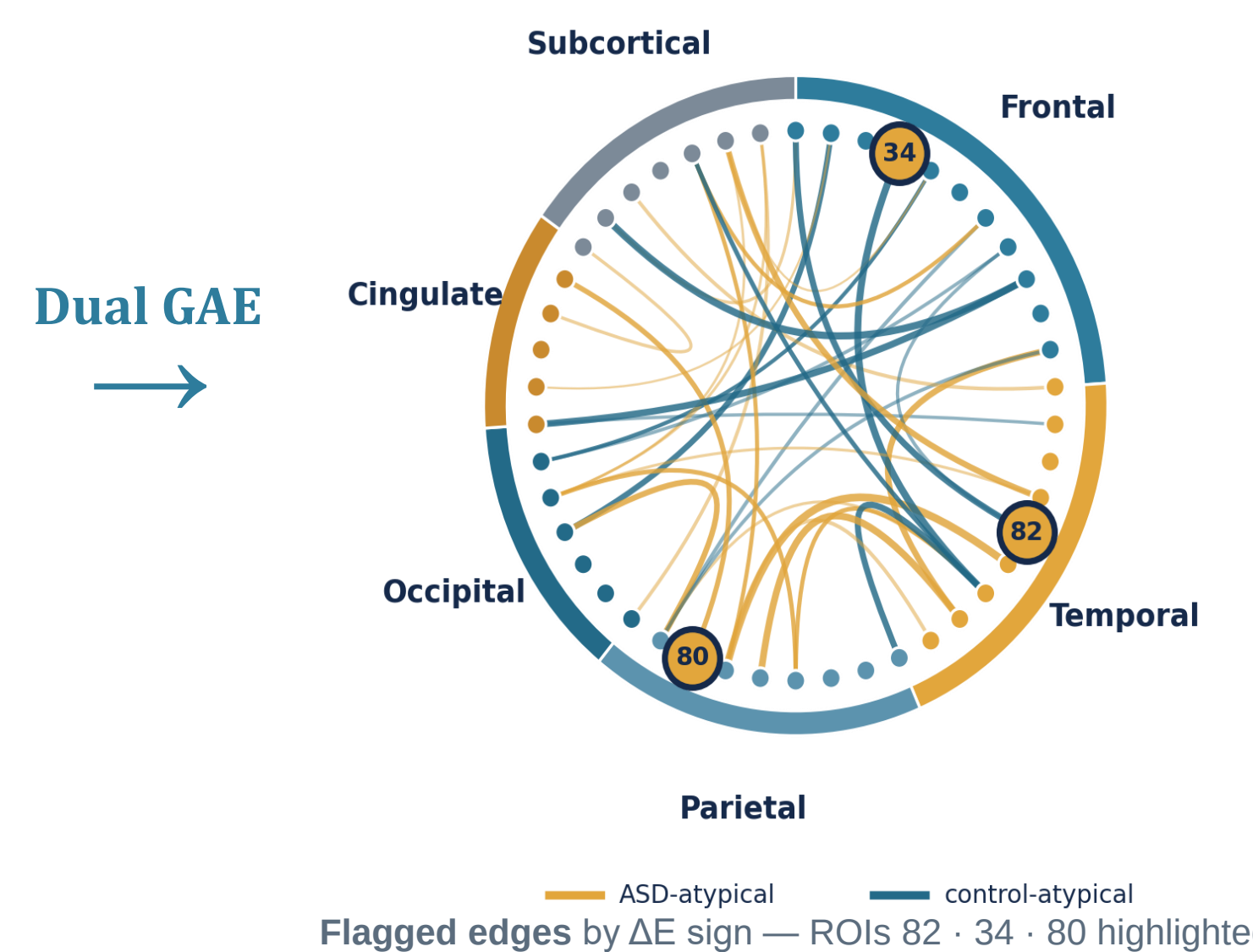


ΔE matrix — each cell an edge; warm = ASD-atypical, cool = control-atypical.

From brain graph to flagged connections



The connectome $G = (X, A)$ — 111 ROIs as nodes



Flagged edges by ΔE sign — ROIs 82 · 34 · 80 highlighted

09 What the model looks at

The three most consistently ranked ROIs across folds — all independently implicated in ASD.

#1

Left inferior temporal gyrus

temporo-occipital part · node 82

Atypical gray-matter properties reported in low- and high-functioning ASD. [Cai et al., 2018]

#2

Right angular gyrus

node 34

Altered functional connectivity in ASD relative to controls. [Xu et al., 2019]

#3

Left superior parietal lobule

node 80

Differential activation during spatial and attentional tasks. [Hamilton, 2013]

Code on GitHub



github.com/GeorgiaKp/interpretable_dual_gae_fmri